

Ahmed Aly

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EDUCATION

Master of Science in Computer Vision

Aug. 2024 – May 2026

Mohamed bin Zayed University of Artificial Intelligence

Abu Dhabi, UAE

- **Thesis:** Training and benchmarking a vision-language foundation model for large-scale echocardiography data.
- Developed a multimodal model learning joint representations from echocardiography videos and text supervision.
- Speaker at the MICCAI 2026 Winter School, delivering a tutorial on foundation models for cardiac ultrasound.
- Co-organized the MICCAI HECKTOR 2025 Challenge, designing evaluation metrics and protocols.
- Managed 36 TB of echocardiography video data for scalable foundation model training.

Bachelor of Science in Applied Mathematics and Statistics

Aug. 2016 – May 2020

Khalifa University

Abu Dhabi, UAE

- **Thesis:** Optimization of global minima in nonlinear physical systems using metaheuristic statistical models.

EXPERIENCE

Machine Learning Engineer, Intern

May 2025 – July 2025

Labib AI

Abu Dhabi, UAE

- Developed a real-time multimodal AI system integrating vision and speech models for surgical scene understanding, enabling context-aware decision support.
- Optimized low-latency inference pipelines for real-time deployment on GPU-based systems.

Computer Vision Engineer

Jan 2023 – August 2024

Koralyze AI

Abu Dhabi, UAE

- Co-founded a football analytics company specializing in the generation of tracking and event data directly extracted from football broadcast footage.
- Analyzed event sequences and player trajectories over time for downstream football analytics.
- Coordinated and presented at a booth during GITEX Expand North Star 2023.

Mathematics Teacher

Aug 2020 – May 2024

Ministry of Education

Sharjah, UAE

- Instructed high school students in AP Calculus AB and BC

RESEARCH PROJECTS

CardioBench: Do Echocardiography Foundation Models Generalize Beyond the Lab?

[preprint](#)

- Designed a benchmark protocol to evaluate out-of-distribution generalization of vision foundation models across views and acquisition settings.

SynSpineMS: Multiple Sclerosis Spinal Cord Lesion Detection from MultiSequence MRIs

[MICCAI challenge](#)

- Led and coordinated our team's participation in the MICCAI MS-Multi-Spine challenge and submission packaging, finished 2nd overall.

Language and Planning in Robotic Navigation

[LM4Plan, AAI 2025](#)

- Designed a Vision-Language Navigation system enabling autonomous agents to interpret natural language instructions and perform multi-step planning in indoor robotic environments.
- Implemented and evaluated language-grounded planning and reasoning across multilingual (Arabic/English) instruction settings.

SKILLS

Optimization: Fine-tuning large models, distributed training, DeepSpeed, large-scale data pipelines.

ML Frameworks: PyTorch, Hugging Face Transformers, PyTorch Geometric, OpenCV, scikit-learn.

Systems & Infrastructure: Docker, GPU-based training, scalable preprocessing pipelines.

Programming: Python (primary), C++ (performance-oriented components).

Communication: English (IELTS 8.0), Arabic (Native).